

HW2 - Stochastic Methods for Industrial Engineers

Submission: July 8, 2026

1 True or False?

For each of the following claims, determine if they are true or false. If you think they are true - prove them. If you think they are false - find a counterexample. Explain your answers clearly.

- a) In an irreducible and infinite Markov chain, if the state is null recurrent, then it is impossible to find a stationary distribution π that satisfies $\pi P = \pi$ and $\sum_i \pi_i = 1$.

- (a) True. If we could find such a stationary distribution, and we initialize the chain with it, then it must be the distribution for all time steps. Since $\sum_i \pi_i = 1$ then we must have states with $\pi_i > 0$. However, for every state i with $\pi_i > 0$ we must have a finite expected return time, otherwise this state cannot occur a fraction π_i of the time. The formal proof goes a bit beyond the tools we learned in class. If π is a stationary distribution, then

$$\mathbb{E}_i[\tau_i] = \frac{1}{\pi_i} < \infty,$$

where

$$\tau_i = \inf\{n \geq 1 : X_n = i\}.$$

Thus every state must be positive recurrent. A null recurrent state satisfies

$$\mathbb{P}_i(\tau_i < \infty) = 1, \quad \mathbb{E}_i[\tau_i] = \infty.$$

Therefore, an irreducible null recurrent chain cannot possess a stationary probability distribution satisfying

$$\pi P = \pi, \quad \sum_i \pi_i = 1.$$

- b) There exists a finite Markov chain that has infinitely many distinct stationary distributions.

- (a) True. Consider the finite Markov chain with transition matrix

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both states are absorbing. For every $\alpha \in [0, 1]$, the probability vector

$$\pi^{(\alpha)} = (\alpha, 1 - \alpha)$$

satisfies

$$\pi^{(\alpha)} P = \pi^{(\alpha)}.$$

Since there are infinitely many choices of α , this finite chain has infinitely many distinct stationary distributions.

- c) Let X_n and Y_n be two independent realizations of the exact same irreducible, finite-state Markov chain. In this case, it is guaranteed that those two chains will eventually be in the same state at the exact same time (i.e., $\mathbb{P}(\exists n \geq 0 \text{ such that } X_n = Y_n) = 1$).

- (a) False. Consider the irreducible two-state chain

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Let

$$X_0 = 0, \quad Y_0 = 1.$$

The chains then evolve deterministically:

$$X_n = \begin{cases} 0, & n \text{ even,} \\ 1, & n \text{ odd,} \end{cases} \quad Y_n = \begin{cases} 1, & n \text{ even,} \\ 0, & n \text{ odd.} \end{cases}$$

Hence

$$X_n \neq Y_n \quad \text{for every } n \geq 0,$$

and therefore

$$\mathbb{P}(\exists n \geq 0 : X_n = Y_n) = 0.$$

- d) If an irreducible Markov chain possesses a unique stationary distribution π , it is not guaranteed that $\lim_{n \rightarrow \infty} P_{ij}^{(n)}$ exists.

- (a) True. Again consider

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This chain is irreducible and has the unique stationary distribution

$$\pi = \left(\frac{1}{2}, \frac{1}{2} \right).$$

However,

$$P^{2n} = I, \quad P^{2n+1} = P.$$

For example,

$$P_{00}^{(2n)} = 1, \quad P_{00}^{(2n+1)} = 0.$$

Thus $\lim_{n \rightarrow \infty} P_{00}^{(n)}$ does not exist. Uniqueness of the stationary distribution does not imply convergence of the transition probabilities; aperiodicity is also needed.

e) Consider the Metropolis algorithm with a uniform target distribution $\pi_i = \pi_j$ over a finite, connected graph G . Suppose we modify the Metropolis transition probabilities such that:

(a) For neighboring nodes i, j , with degrees d_i, d_j ,

$$P_{i,j} = \frac{1}{\max(d_i, d_j)}$$

(b) For non-neighbors i, j , $P_{i,j} = 0$.

(c) Self-loop:

$$P_{i,i} = 1 - \sum_{j \neq i} P_{i,j}.$$

Then the resulting Markov chain is ergodic with a uniform stationary distribution.

i. **False. The chain need not be aperiodic. Consider a graph consisting of two vertices and an edge between them. Then $d_1 = d_2 = 1$, so the transition matrix is**

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The chain has period 2. Hence it is not ergodic by definition.

f) In an absorbing Markov chain, if state i is transient and state j is absorbing, then the single-step transition probability must be positive, $P_{i,j} > 0$.

(a) **False. Consider the Markov chain with the transition matrix $P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. The first state is transient, but has zero probability to move directly to the absorbing third state.**

g) Let X_n be a Markov chain. For a state i , define $\tau_i = \inf\{n \geq 1 : X_n = i\}$. If $\mathbb{P}(\tau_i < \infty \mid X_0 = i) = 1$, then $\mathbb{E}[\tau_i \mid X_0 = i] < \infty$.

(a) **False. We saw a counterexample example in slide titled “Example: Recurrence” in class.**

h) Let X_n be a time-homogeneous Markov chain with a finite state space S and transition matrix P . If the chain is irreducible and ergodic, then its unique stationary distribution π satisfies:

$$\pi_i P_{i,j} = \pi_j P_{j,i} \quad \forall i, j \in S$$

(a) **False. A counterexample is the Markov chain with transition matrix**

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}.$$

This chain is irreducible and aperiodic, hence ergodic. Since $1P = 1$ then its unique stationary distribution is $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. However,

$$\pi_0 P_{01} = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6},$$

while

$$\pi_1 P_{10} = \frac{1}{3} \cdot 0 = 0.$$

Thus,

$$\pi_0 P_{01} \neq \pi_1 P_{10},$$

so the $\pi_i P_{i,j} = \pi_j P_{j,i}$ equations do not hold.

- i) Let $M = \langle S, A, P, R, \gamma \rangle$ be an infinite-horizon Markov Decision Process (MDP) with a discount factor $0 < \gamma < 1$. Let $\Psi : S \rightarrow \mathbb{R}$ be some (unknown) bounded function. Suppose that the reward function of an MDP M is modified to create a new MDP M' , such that: $R'(s, a, s') = R(s, a, s') + \Psi(s) - \gamma\Psi(s')$. The optimal policies for M and M' are identical.

(a) Let

$$G = \sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1})$$

denote the discounted return in the original MDP. In the modified MDP,

$$R'(s, a, s') = R(s, a, s') + \Psi(s) - \gamma\Psi(s'),$$

so the new return is

$$G' = \sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t, s_{t+1}) + \Psi(s_t) - \gamma\Psi(s_{t+1})) =$$

$$G + \sum_{t=0}^{\infty} (\gamma^t \Psi(s_t) - \gamma^{t+1} \Psi(s_{t+1}))$$

The second sum is telescoping:

$$\sum_{t=0}^N (\gamma^t \Psi(s_t) - \gamma^{t+1} \Psi(s_{t+1})) = \Psi(s_0) - \gamma^{N+1} \Psi(s_{N+1}).$$

Since $0 < \gamma < 1$ and Ψ is bounded,

$$\lim_{N \rightarrow \infty} \gamma^{N+1} \Psi(s_{N+1}) = 0.$$

Hence,

$$G' = G + \Psi(s_0).$$

For a fixed initial state s_0 , the term $\Psi(s_0)$ is independent of the actions taken. Therefore, every policy's expected return is shifted by the same constant, so the ordering of policies is unchanged. Consequently, the optimal policies of M and M' are identical.

- j) Let X_n be a finite ergodic Markov chain with at least two distinct states, with a unique stationary distribution π . Suppose X_1 is distributed like π . Then, it is always true that the sequence converges in distribution ($X_n \xrightarrow{d} X$ where X is distributed like π), but it is strictly impossible for the sequence X_n to converge in probability to any random variable.

- (a) True. Since $X_1 \sim \pi$ and $\pi P = \pi$, stationarity gives $X_n \sim \pi$ for every $n \geq 1$. Consequently, $X_n \xrightarrow{d} X$ for $X \sim \pi$. Suppose, for contradiction, that X_n converges in probability to some state-valued random variable X . Since the state space is finite and therefore discrete,

$$\mathbb{P}(X_n \neq X) \longrightarrow 0.$$

By the union bound,

$$\mathbb{P}(X_{n+1} \neq X_n) \leq \mathbb{P}(X_{n+1} \neq X) + \mathbb{P}(X_n \neq X) \longrightarrow 0.$$

On the other hand, because the chain is stationary,

$$\mathbb{P}(X_{n+1} \neq X_n) = \sum_i \pi_i (1 - P_{ii}),$$

which is constant in n . Therefore, it must equal zero:

$$\sum_i \pi_i (1 - P_{ii}) = 0.$$

For a finite irreducible chain, $\pi_i > 0$ for every i , so this implies

$$P_{ii} = 1 \quad \text{for every } i.$$

That would make every state absorbing, contradicting irreducibility when the chain has at least two states. Hence X_n cannot converge in probability to any random variable.

- k) Let $(X_n)_{n=0}^T$ be a finite irreducible homogeneous Markov chain with transition matrix P and stationary distribution π . Define the reversed process

$$Y_n = X_{T-n}, \quad 0 \leq n \leq T.$$

Then $(Y_n)_{n=0}^T$ is also a homogeneous Markov chain.

- (a) False. Consider the two-state Markov chain with transition matrix

$$P = \begin{pmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{pmatrix}$$

and initial condition $X_0 = 0$ with probability 1. The chain is finite, irreducible, and homogeneous. Let $T = 2$. Then $Y_0 = X_2$, $Y_1 = X_1$, $Y_2 = X_0$.

The distributions of X_n are

$$\mu_0 = (1, 0), \quad \mu_1 = (0.9, 0.1), \quad \mu_2 = (0.83, 0.17).$$

Hence,

$$\Pr(Y_1 = 0 \mid Y_0 = 0) = \Pr(X_1 = 0 \mid X_2 = 0) = \frac{0.9 \cdot 0.9}{0.83} = \frac{0.81}{0.83} \approx 0.976,$$

whereas

$$\Pr(Y_2 = 0 \mid Y_1 = 0) = \Pr(X_0 = 0 \mid X_1 = 0) = 1.$$

Thus the transition probabilities of the reversed process depend on time, so (Y_n) is not a homogeneous Markov chain.

In class, there was the extra condition that X_0 is distributed like π . Then $\Pr(X_t = i) = \pi_i$, $\forall t \geq 0$. For any $0 \leq n < T$,

$$\Pr(Y_{n+1} = j \mid Y_n = i) = \Pr(X_{T-n-1} = j \mid X_{T-n} = i) = \frac{\Pr(X_{T-n} = i \mid X_{T-n-1} = j) \Pr(X_{T-n-1} = j)}{\Pr(X_{T-n} = i)} = \frac{P_{ji} \pi_j}{\pi_i}.$$

Define $\tilde{P}_{ij} = \frac{\pi_j P_{ji}}{\pi_i}$. Since $\sum_j \tilde{P}_{ij} = \frac{1}{\pi_i} \sum_j \pi_j P_{ji} = \frac{(\pi P)_i}{\pi_i} = 1$, the matrix $\tilde{P}$ is stochastic.

Therefore, $\Pr(Y_{n+1} = j \mid Y_n = i) = \tilde{P}_{ij}$, which is independent of n . Hence (Y_n) is a homogeneous Markov chain with transition matrix $\tilde{P}_{ij}$.

1) In an irreducible Markov chain, if one state is aperiodic, then every state is aperiodic

(a) True. For a state i , its period is

$$d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}.$$

In an irreducible Markov chain, all states communicate, and communicating states have the same period. Indeed, if i and j communicate, there exist $r, s \geq 1$ such that

$$P_{ij}^{(r)} > 0, \quad P_{ji}^{(s)} > 0.$$

Every return path from j to itself can be combined with paths from i to j and from j to i , relating the possible return times of the two states. This implies

$$d(i) = d(j).$$

Therefore, if one state is aperiodic, so that $d(i) = 1$, then every state has period 1 and is aperiodic.